# Tautological instances are valid

The Open Logic Project contributors

openlogic:normal-modal-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

*Editorial note.* The counted statement is the author proposition labelled prop:valid-taut (1364-1366); the proof span 1368-1377 is the author proof environment that immediately follows it.

*Editorial note.* External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

*Editorial scope.* One original result of the Open Logic Project unit normal-modal-logic/syntax-and-semantics, counted once: tautological instances are valid. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

*Allowed standard prerequisites (editorial).*

- Tautological instance. A tautological instance is a sentence obtained from a propositional tautology by uniformly substituting modal sentences D_1, ..., D_n for the propositional variables p_1, ..., p_n.. It is the class of sentences the counted proposition claims valid (1364-1366).
- Tautologies under valuations. A propositional tautology is true under every valuation; the unit proves this as its own lemma.. It delivers the contradiction once the mimicking valuation is defined (1374-1377).

*Source reference.* Original lemma lem-valid-taut: sources/olp-normal-modal-logic-syntax-and-semantics/pinned-original-source.tex, source lines 1252–1252. The author proof cites the unit's lemma that tautologies hold under every valuation. The load-bearing work is carried by that separately proved lemma of the same unit; the counted proof retains its own valuation construction in full.

**Proposition 0.1.** *All tautological instances are valid.*

*Proof.* Contrapositively, suppose $!A$ is such that $/ \models M[!A[!D_1[p_1/,] \ldots, !D_n[p_n//]]][w]$, for some model $M \models$ and world $w$. Define an assignment $\sigma v$ such that $\sigma v(p_i) = \top$ if and only if $M \models !D_i[[]w]$ (and $\sigma v$ assigns arbitrary values to $q \notin \{p_1, \ldots, p_n\}$). Then by lem-valid-taut, $/ \models v\,[!A]$, so $!A$ is not a tautology. □

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